

The “First Breaks” identification algorithm

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About



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Over 10 years of experience in R&D, machine learning and implementation of innovative technologies, with expertise in computer vision, MLOps, and neural networks. Recognized with the "Best Engineer of Russia" award in 2021. Combines practical knowledge in natural science and chemical industry with strong skills in machine learning, computer vision, and data analysis to drive efficiency and optimize production.

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Project code repository: https://github.com/DimYun/test-task_2

About: terminology and problem overview

First Break - initial specific waves that are the first to be detected on a seismogram (first breaks, first arrivals).
Traditionally - performed manually, need to be automate.

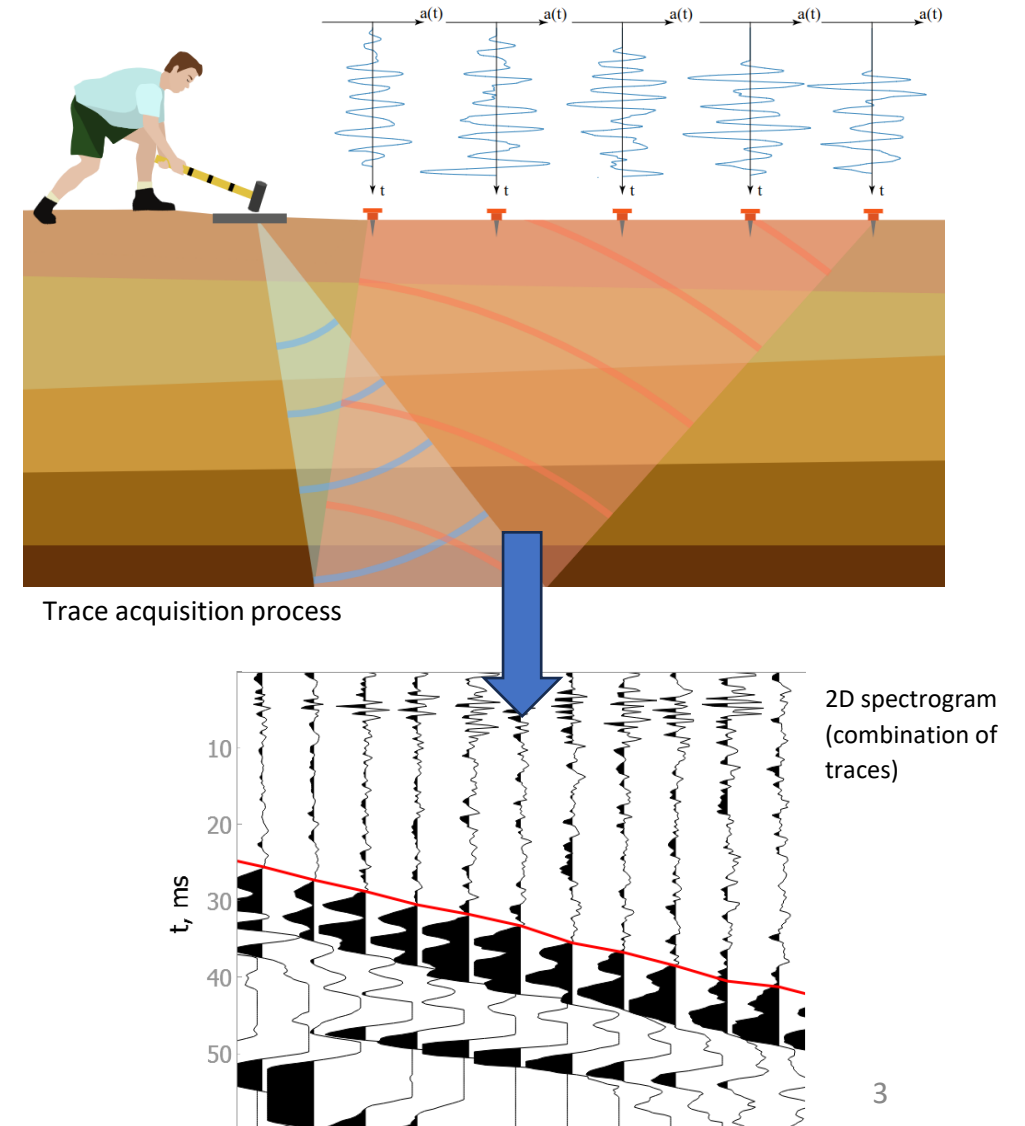
Trace - 1D seismic waves or seismogram.

2D spectrogram - seismic images, set of traces from receivers' batch since the data on adjacent seismic traces have similarities in the features of the wave field.

Problems:

- The number of picks reaches hundreds of thousands in the field data with a lot of noises
- Existing analytical methods allow you to automate picking only on high-quality data with a high signal / noise ratio

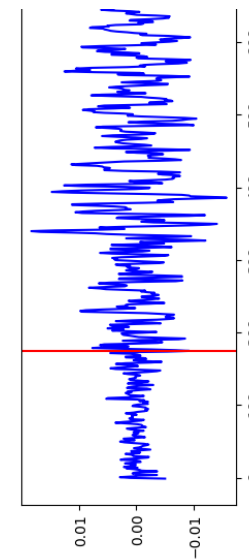
Possible solution: more robust algorithm, based on statistical knowledge or neural network to pick the first breaks.



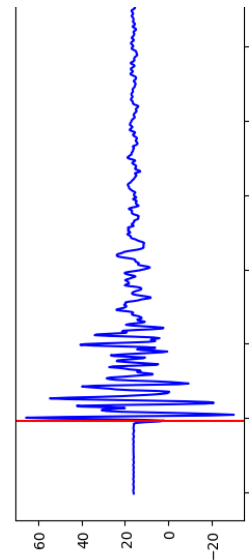
Exploratory Data Analyses

1. Efficient upload of the data from hdf5 files: more than 39 Gb, apply **np.float16** for traces, **np.int16** for SAMP_RATE and SPARE1
2. Check dimension of data and misses: **trace lengths may be different** (751, 1501, etc. points); trace present in condense form (the x-axes data need to be transform with SAMP_RATE to feat the ms, or we can transform SPARE1 instead); **9 831 242 traces** in total, **37.56% of missed first breaks**
3. According to low missing rate **we can skip first break bad values if we working with traces**, or we can **fill them with average values from neighbors for 2D spectrogram**
4. Use **REC_X, REC_Y and REC_HT (to ensure correct choices)** to separate receivers to plot 2D spectrograms: **13113 2D spectrogram in total**. For total data process we add number of datasours: 0 - Brunswick, 1 - Halfmile3D, 2- Lator, 3 - preprocessed
5. According to visual analyses, **some 1D data have inaccurate first break indication or 'hard-to-tell' values**. We need to develop a procedure to filter them according to select algorithm
6. Analyse different data representation according to select type of algorithm for first break calculation (describe later)

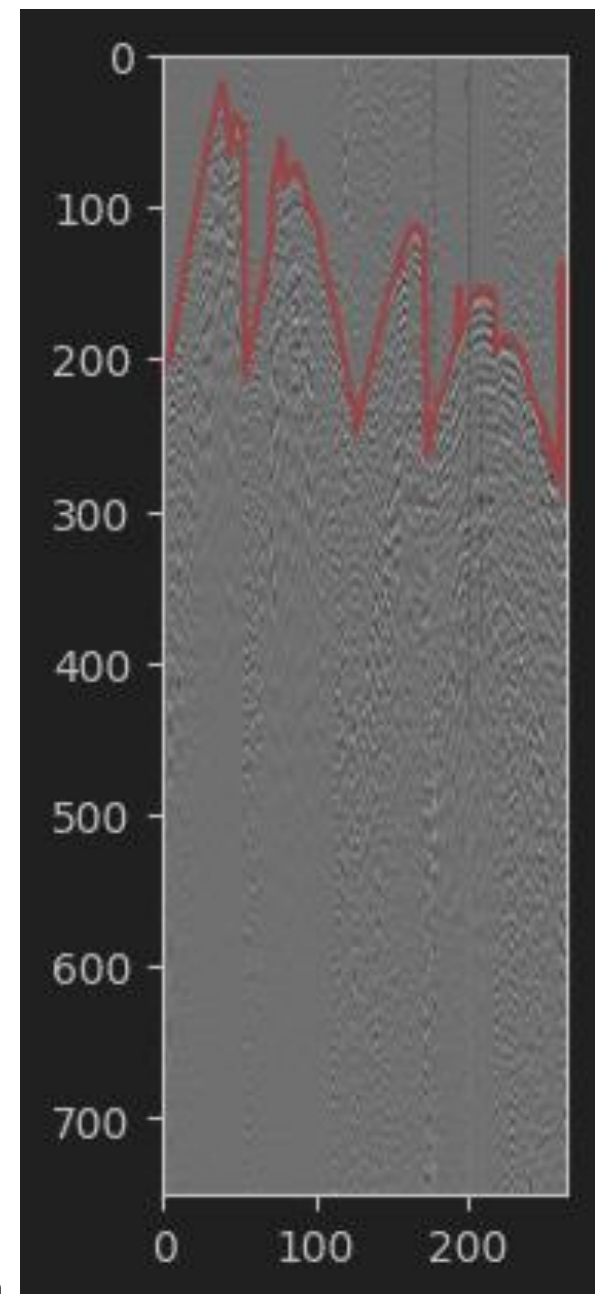
Report with code: notebooks/EDA.ipynb



Plot of single trace with first break (red, hard to identify or error)



Plot of single trace with first break (red, easy to identify, valid)



2D spectrogram example, first breaks show in red (after fill with average values)

Possible solutions*

No	Type and Name	Pros	Cons
1	Analytical. Fractal-based for single trace, search change in dimension ("divider-method")	SNR** ≥ 1.2 ($\approx 3\text{dB}$) Can work with single trace	Slow 1996y. - old Can't work with multi-dimensional data Try to simulate "shape" of signal as human brain - CNN more effective
2	Analytical. Multi-window real-time detection, based on amplitude analyses.	Real-time 2006y. - relatively new Can work with single trace	SNR ≥ 3 ($\approx 10\text{dB}$) Thresholds setup manually, according to expecting SNR
3	Probabilistic. Based on information theory.	2017y. - new Use nature of signals	Slow SNR ≥ 1.5 ($\approx 5\text{dB}$, but pretty unclear) Can't work with single trace
4	Deep Learn. A regression approach, based on bidirectional long short-term memory (BiLSTM) neural network by learning the implicit Eikonal equation of 3D inhomogeneous media with rugged topography in the target region.	2024y. - new Use nature of signals Address to low SNR	Slow Based on multidimensional data LSTM neural network - old architecture Didn't show real SNR
5	Analytical (entropy-based). Modified Coppens's method, use Hausdorff measures.	2010y. - relatively new Can work with single-trace	Manually-tuned parameters Didn't show real SNR Slow
6	Deep Learn. Based on energy-amplitude form of signals with image analysis with UNet model	2025y. - new Self loss function with regularization Great possibility of image combination Can work with single trace	Manually-tuned parameters No SNR estimation No model speed estimation

*based on literature review

** SNR – signal to noise ratio

Selected solution

Computer Vision – most interesting and fresh!

Based on literature review the **Deep Learning methods overcome classical approaches** with

- greater possibility to shape signals (based on their nature or shapes)
- work with single and multidimensional data
- more real data applicability
- greater versatility

According to found information, computer vision approach with UNet model and self-written loss function looks promising. Based on other literature information we can improve it further with different signal representation, adding boundary-specific function for loss (e.g. Hausdorff) or use windowed transformation of 1D traces.

Selected solution: proposed CV model

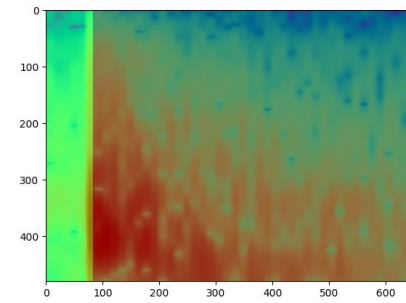
According to describe path I propose next model:

1. **UNet base**: exhibit superior learning capacity by leveraging comprehensive frame feature information
2. **MobileNetV4** backbone: time-effective calculation with possibility to implement on CPU or edge-devices
3. **Combined loss** $L = 0.2 * BCE + 0.4 * Dice + 0.4 * BoundaryIoU$: this function incorporates Binary Cross-Entropy (BCE) loss to handle pixel-wise classification, Dice loss to optimize the overlap between predicted and ground truth masks, and Boundary IoU loss to specifically refine the accuracy of object boundaries (is proven to be more stable, than Hausdorff distance and allows to train model even without the other losses in equation). The specific weights for each loss component were empirically determined to optimize performance
4. **IoU metric** for better explainability with batch visualization and fast implementation
5. **Simple augmentations** to preserve high frequency information and small patterns with high resolutions: HorizontalFlip, CoarseDropout, OneOf(GaussianBlur, ISONoise, ImageCompression)
6. Train procedure: fast (≤ 13 epochs) with small image (≤ 256 pixels) to feat into the deadline

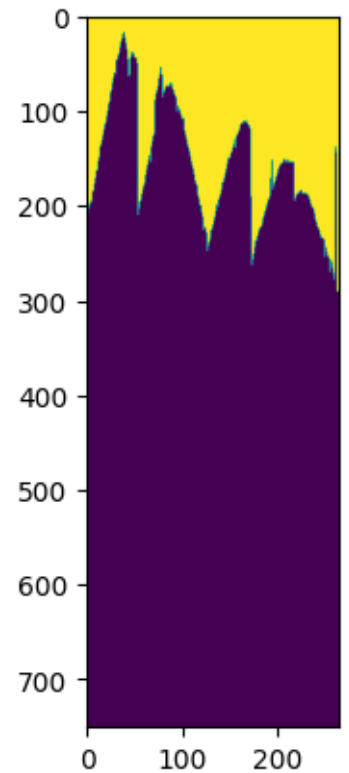
Preparing datasets

To test whole possibilities of proposed NN model **there are 2 possible cases:**

1. Analyze **separate traces** - images build from 1D trace with binary mask of noise data before first break:
 1. Drop invalid first break data (traces with 0 or -1 first break value)
 2. Draw binary mask of noise data before first break as rectangle
 3. Randomly **select 100 000 traces** with known break values and split it to **70% train, 20% validate and 10% test data. Evaluate on randomly selected 100 data** with mean error and std calculation to test accuracy and repeatability
2. Analyze **2D spectrogram** of each receiver:
 1. Replace 0 and -1 values of first breaks with average value of neighbors
 2. Draw binary mask of noise data before first break with complex forms
 3. **Select all 13 113 images** and split them on **70% train, 20% validate and 10% test data. Evaluate on randomly selected 100 data** with mean error and std calculation to test accuracy and repeatability



Example of processed single trace with binary mask



Example of 2D spectrogram binary mask

Results Validation

According to selected algorithm and data structure I used:

- **Randomly selected 100 images** from the whole dataset to evaluate metrics
- **Dataset-specific calculation of first break time** (for spectra set 1 and 2 I used recalculation through spectra image wight, for other sets - direct calculation from masks)
- **Augmentation and image shapes were settled equally** for each spectra set (256x256 pixels), except for 2nd set, where images are much smaller (64x64 pixels)
- For model speed estimation I select best approach, convert model to ONNX file and **calculate FPS with warmup and 2000 parallel calculation with averaging** on NVIDIA GeForce RTX 4050 Laptop GPU (6Gb) and Intel(R) Core(TM) Ultra 7 155H CPU

Results Overview - spectra set No 1

Technique: use *matplotlib.pyplot.specgram* to implement FFT on windowed trace 1D data

Pros:

- Fourie transform to highlight interval spectra
- Colored image represent different information about trace
- A lot of tunable parameters

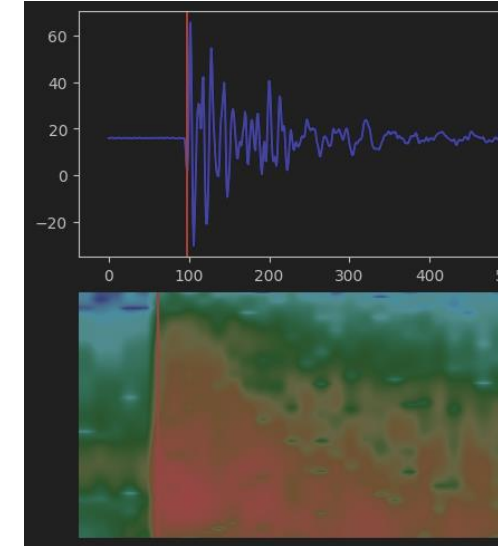
Cons:

- Manual tune a lot of parameters
- Windowed transformation with overlapping
- Low resolution
- Relative first break calculation for image, need to recalculate

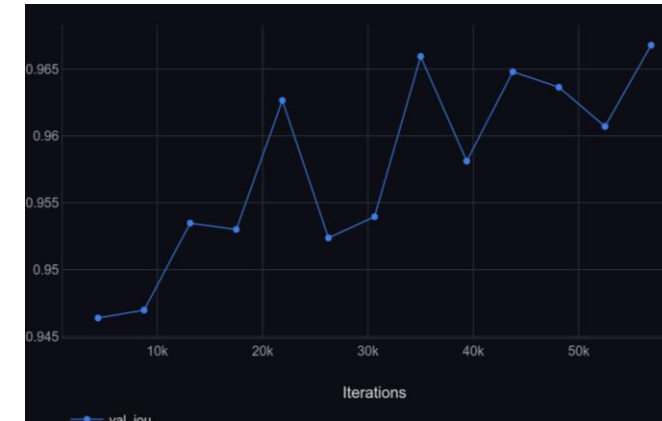
Results*:

- IoU > 0.964
- Good visual determination of first break
- Good visual mask matching
- Average relative error 2.44 % with 3.57 % std – best model
- Can work in teal time: 39 fps on CPU, 188 fps on GPU

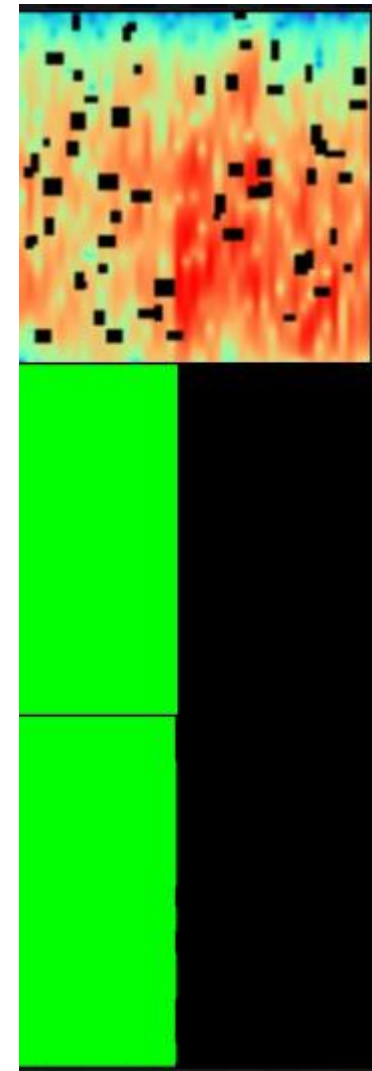
*report in <https://app.clear.ml>, need registration



Example of processed single trace



IoU of validation data thought the training



Example of batch image after augmentations with real mask (in the middle) and predicted mask (on the bottom)

Results Overview - spectra set No 2

Technique: use self-written windowed FFT transformation to create low-res images from trace 1D data

Pros:

- Fourie transform to highlight interval spectra without overlapping
- Fast training and process due to low resolution
- Few tune parameters (window size)

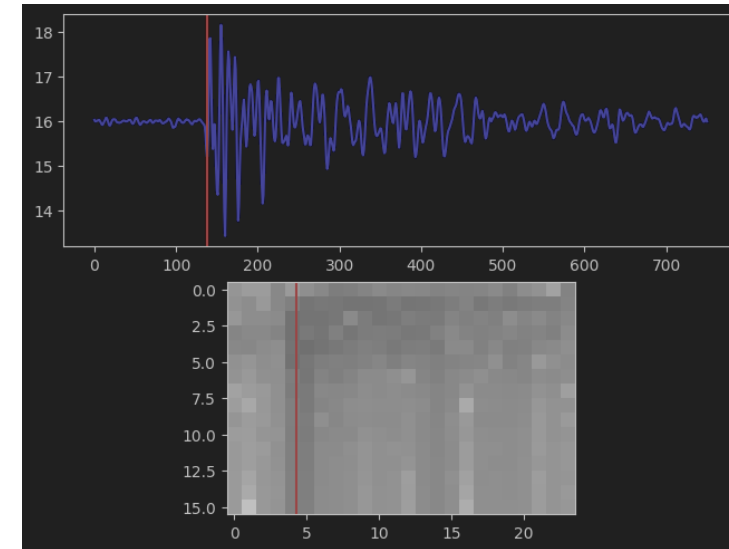
Cons:

- Averaged results with no exact first break place (can predict only window)
- Low resolution

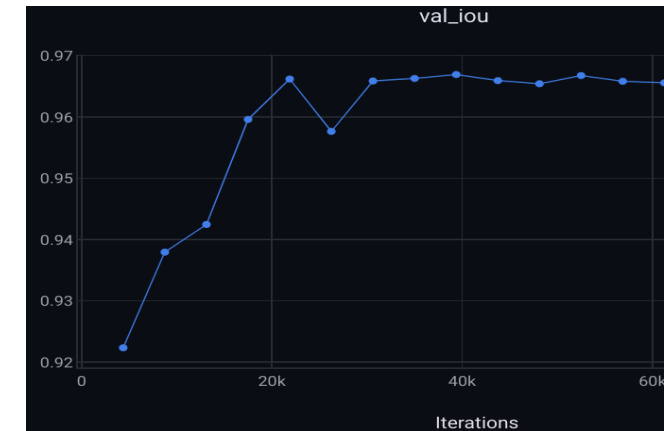
Results*:

- IoU > 0.967
- Reliable visual determination of first break
- Fast training and data process
- Good visual mask matching
- Average relative error 13.21 % with 37.67 % std.

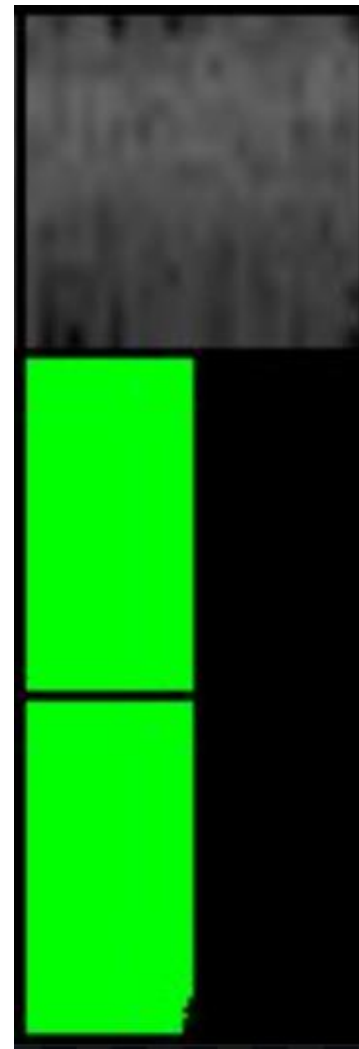
*report in <https://app.clear.ml>, need registration



Example of processed single trace



IoU of validation data thought the training



Example of batch image after augmentations with real mask (in the middle) and predicted mask (on the bottom)

Results Overview - spectra No 3

Technique: use self-written simple stacking to create image from repeated 1D trace

Pros:

- Simple and fast data preparation with direct image structure
- No tunable parameters
- Direct first break value calculation

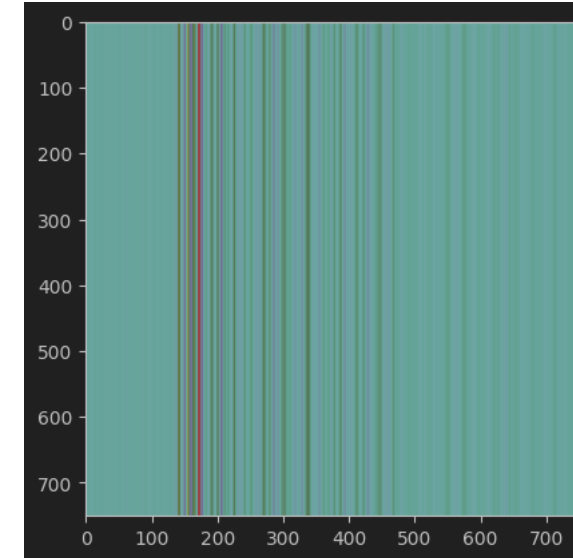
Cons:

- Small patterns on the image with high frequency – hard for Unet
- Relatively big initial resolution of image
- Slow training

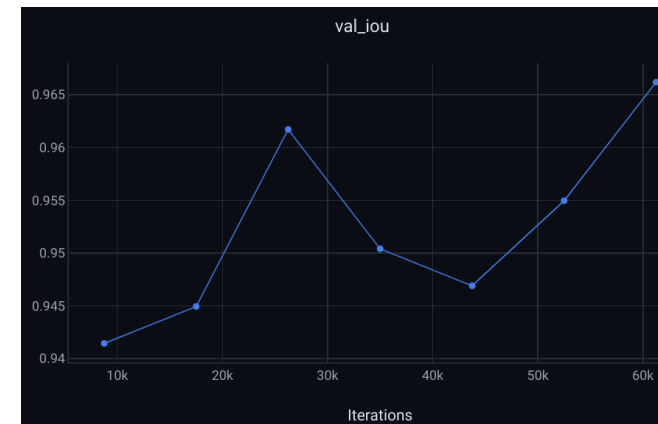
Results*:

- IoU > 0.965
- Good visual determination of first break
- Good visual mask matching
- Average relative error 14.04 % with 80.80 % std.

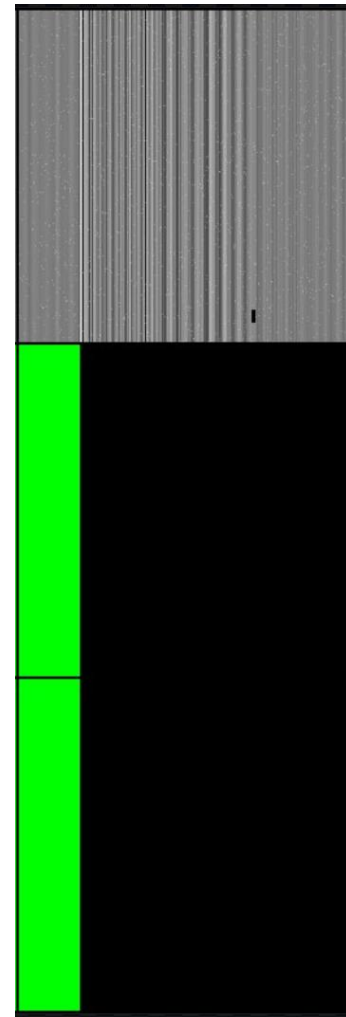
*report in <https://app.clear.ml>, need registration



Example of processed single trace



IoU of validation data throughout the training



Example of batch image after augmentations with real mask (in the middle) and predicted mask (on the bottom)

Results Overview - spectra No 4

Technique: self-developed technique to create image from 1D trace with gauss approximation of each point of trace, based on std from 64 neighbor points and feat into 0-255 range

Pros:

- Gauss approximation of each point of trace, which can be good threshold for noise data (noise more likely be normal distributed than the signal)
- Form-reach images
- One tunable parameter (window for std calculation for Gauss approximation)
- Direct first break time calculation

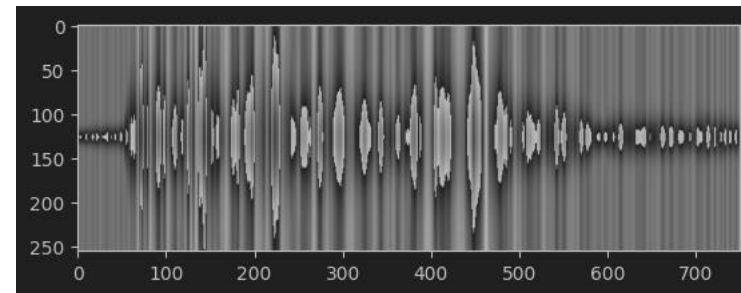
Cons:

- Manual tune parameter
- A lot of fine structures on the image – may be hard for UNet
- Relatively big initial resolution of image

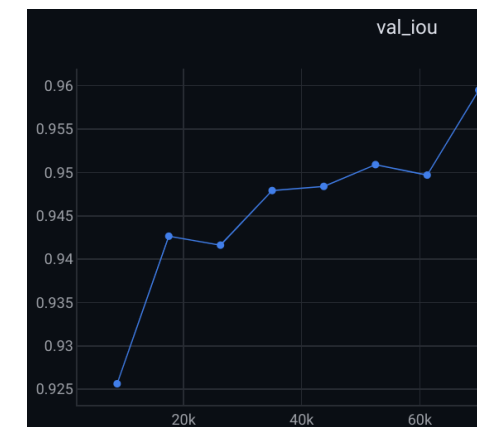
Results*:

- IoU > 0.964
- Good visual determination of first break
- Preserve some artifacts after 1st epoch - interesting
- Average relative error 6.92 % with 31.69 % std – need more epochs

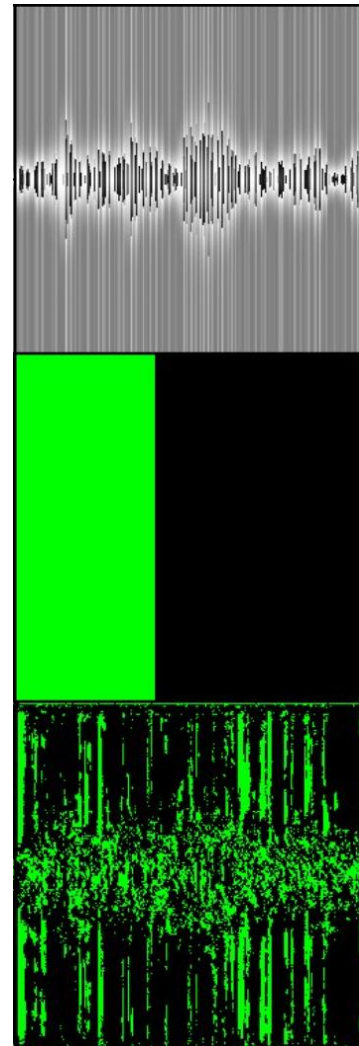
*report in <https://app.clear.ml>, need registration



Example of processed single trace



IoU of validation data throughout the training



Example of batch image after augmentations with real mask (in the middle) and predicted mask (on the bottom) after 1st epoch

Results Overview - spectra No 5

Technique: use classical 2D spectrogram from each receiver series
(simple stacking traces for one receiver)

Pros:

- Can be more information due to single receiver nature
- Form-reach images
- No tunable parameter
- Direct first break time calculations
- Batch processing (save time)

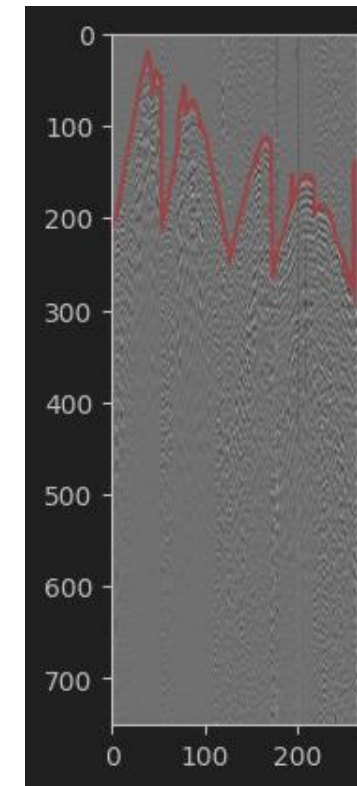
Cons:

- Complicate mask of the image
- A lot of fine structures on the image – may be hard for UNet
- Relatively big initial resolution of image – slow training

Results*:

- IoU > 0.965
- Good visual mask matching and IoU score
- Average relative error 33.60 % with 24.02 % std.

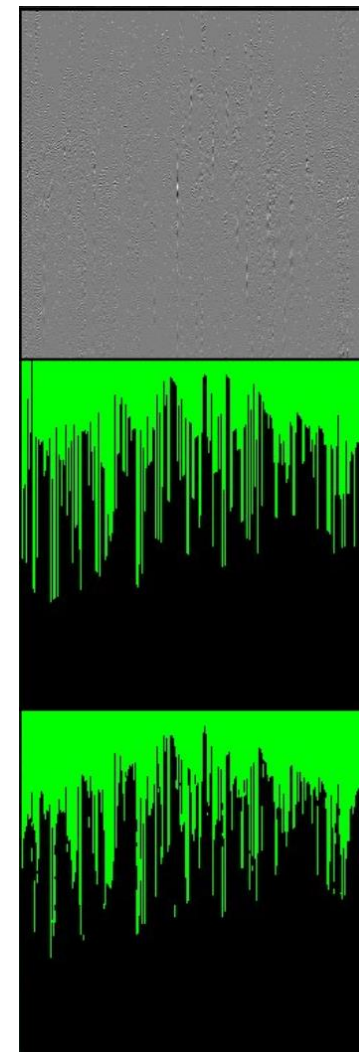
*report in <https://app.clear.ml>, need registration



Example of processed single trace



IoU of validation data throughout the training



Example of batch image after augmentations with real mask (in the middle) and predicted mask (on the bottom)

Conclusions*

1. A first break detection algorithm was successfully developed, achieving an average relative error of 2.44% (with a standard deviation of 3.57%).
2. A computer vision approach was selected for first break detection, based on a literature review.
3. Five graphical representation methods were studied: four using single-trace images and one using 2D spectrograms from each receiver. All methods were implemented and tested.
4. The windowed Fourier approximation (spectra set No. 1) proved to be the optimal solution, enabling real-time application on CPU devices at 39 fps.
5. A self-developed Gaussian approximation approach for separating noise structures from signals yielded acceptable results, with an average relative error of 6.92% (and a standard deviation of 31.69%). The high standard deviation suggests that further training is needed to improve stability.
6. A neural network pipeline with a custom loss function was justified, implemented, and adapted for efficient experimentation.
7. A combined loss function with border attention was implemented and demonstrated effectiveness for precise mask prediction.
8. All approaches achieved an Intersection over Union (IoU) greater than 0.95 and an error less than 15%, except for the 2D spectrogram method, which had a 33% error due to limited training time and small image resolution.
9. All proposed approaches can be used to identify a window containing the first break, enabling the application of analytical search algorithms or detailed signal structure analysis.
10. The proposed solution is capable of real-time calculations on CPU devices (39 fps) and achieves 188 fps on GPU.

***Project code repository:** https://github.com/DimYun/test-task_2

Limitations

Due to time constraints, the following limitations were encountered:

- **Model Architecture:** Only a UNet architecture with a MobileNetV4 backbone was explored.
- **Training Time:** Models were trained for a limited number of epochs (less than 13 on average), while optimal performance requires significantly more (e.g., >100 epochs).
- **Image Resolution:** Input image resolution was limited to 256x256 pixels (and 64x64 for spectra No. 2). Spectra images should ideally not be compressed by more than 2x to preserve critical information.
- **Loss Function:** A predefined combined loss with boundary IoU was used. Exploration of alternative loss functions, such as Fourier loss and other distance-based losses, is recommended.
- **Dataset Size:** Only a subset (100,000) of the total dataset (9,000,000+ traces) was used for training. Utilizing the full dataset could enhance model generalization.
- **Data Quality:** A thorough visual inspection of data and masks is needed to identify and address potential outliers.

Additionally, limited information on the data's physical nature and experimental setup imposed these constraints:

- **Transform Functions:** While Fourier and Gaussian approximations were studied, Wavelet functions should be considered if they align with the initial signal generation.
- **Spectrogram Aggregation:** 2D spectrograms were combined solely based on receiver XYZ coordinates. Incorporating receiver group information could yield more informative spectrogram representations.
- **Missing Value Fill:** Missing first break values were filled using the average of neighboring values. Alternative imputation methods (e.g., median, distribution-based) should be investigate.
- **Image Format:** JPG images were used for spectra representation to improve processing speed. Higher fidelity formats like PNG are recommended to prevent information loss.

References

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